

## Project overview

Students will conduct a comprehensive virtual screening workflow to identify potential drug candidates for a specific protein target. Each student receives a target protein and a reference active ligand, then follows a structured process incorporating both ligand-based and structure-based virtual screening techniques.

## Learning objectives

By completing this project, students should demonstrate understanding of:

- The complete drug discovery computational pipeline
- Data acquisition and preparation from biological databases
- Molecular filtering techniques and their rationale
- Ligand-based virtual screening methodologies
- Machine learning applications in drug discovery
- Structure-based virtual screening using molecular docking
- Critical analysis and interpretation of computational results

## Detailed project guide

### Step 1: Compound data acquisition

**What students need to know:**

- The structure and organization of the ChEMBL database
- How biological activity data is stored and accessed
- The relationship between protein targets, compounds, and activity data
- The meaning and calculation of pIC50 as a measure of biological activity

**How to perform the analysis:**

1. Use the `chembl_webresource_client` Python package to connect to the ChEMBL database
2. Query the target protein using its UniProt ID to obtain the ChEMBL target ID
3. Retrieve all compounds with reported activity against this target
4. Extract essential information: compound structures (SMILES), IC50 values, and other relevant bioactivity data
5. Calculate pIC50 values from IC50 data using the proper formula for that.

**What to demonstrate:**

- Clear explanation of how the ChEMBL database was queried
- Summary statistics of the retrieved compound set (number of compounds, activity range)
- Visualization of the activity distribution of the compound set

- Research your protein target - what does it do, how does it look and why is it pharmaceutically relevant? Mechanism of action for ligand with known activity to target protein.

## Step 2: Molecular filtering

### What students need to know:

- Principles of drug-likeness and Lipinski's Rule of Five (RO5)
- Common problematic substructures in drug discovery (PAINS)

### How to perform the analysis:

For Lipinski's Rule of Five:

1. Calculate key physicochemical properties for each molecule:
  - Molecular weight (cutoff: 500 Da)
  - LogP (cutoff: 5)
  - Hydrogen bond donors (cutoff: 5)
  - Hydrogen bond acceptors (cutoff: 10)
2. Apply Lipinski's Rule with the flexibility that one violation is acceptable
3. Create property distribution visualizations (histograms, etc)
4. Generate radar plots for representative compounds showing their properties relative to ideal ranges

For PAINS filtering:

1. Screen molecules against established PAINS substructure patterns
2. Identify compounds containing problematic substructures
3. Document specific PAINS patterns found in the excluded compounds

### What to demonstrate:

- Clear rationale for filtering decisions
- Statistics of how many compounds passed/failed each filter
- Specific examples of compounds that failed filtering with explanations:
  - Show molecular structures with highlighted violations
  - Explain why these particular structures are problematic (e.g., specific PAINS substructures and their mechanisms of interference)
- Radar plots showing property distributions before and after filtering (for RO5)

## Step 3: Ligand-based virtual screening

### What students need to know:

- Concept of molecular fingerprints and chemical similarity
- Different types of fingerprints and their strengths/weaknesses
- Similarity metrics and their mathematical basis
- Enrichment as a measure of virtual screening effectiveness

### How to perform the analysis:

1. Generate multiple types of molecular fingerprints for all compounds:
  - MACCS keys (166-bit structural keys)
  - Morgan/ECFP fingerprints (with radii 2, 4, and 6)
  - Topological fingerprints
  - Atom Pairs or other fingerprint types
2. Calculate similarity metrics between each compound and the reference ligand:
  - Tanimoto coefficient (most common)
  - Dice coefficient (alternative metric)
3. Rank compounds by similarity to the reference compound
4. Create enrichment plots:
  - Define active compounds based on a pIC50 threshold
  - Plot percentage of actives found vs. percentage of database screened
  - Calculate metrics like AUC (ROC plot), enrichment factor at 1%, 5%, and 10%

### What to demonstrate:

- Comparison of different fingerprint types' effectiveness
- Analysis of similarity score distributions
- Enrichment plots, ROC curve
- Dependency of entire analysis on the definition of active / inactive (how did you find proper threshold)
- Critical analysis of which fingerprint performed best and why
- Top-ranked compounds by similarity (to ligand with known activity), with visualizations of their structures compared to the reference ligand

## Step 4: Clustering analysis

### What students need to know:

- Principles of clustering as an unsupervised machine learning method
- Different clustering algorithms and their properties
- Methods for determining optimal cluster numbers
- Concept of maximum common substructure (MCS)

### How to perform the analysis:

1. Select appropriate fingerprints from the previous phase
2. Implement clustering methods:
  - K-means clustering (requires pre-specifying number of clusters)
  - Density-based clustering (DBSCAN, automatically determines clusters)
3. Determine optimal parameters:
  - For K-means: use methods like the elbow method, silhouette score
  - For HDBSCAN: determine appropriate params
4. Identify which cluster contains the reference active compound
5. Analyze the compounds that cluster with the reference compound
6. Calculate the maximum common substructure of compounds in this cluster

**What to demonstrate:**

- Visual representation of the clustering (e.g., PCA plot colored by cluster)
- Justification for the selected fingerprint type, clustering method and clustering parameters
- Analysis of cluster containing the reference ligand:
  - Size and composition of the cluster
  - Activity distribution within the cluster
  - Structural features shared by cluster members
- Visualization of the maximum common substructure

**Step 5: Machine learning for activity prediction****What students need to know:**

- Principles of supervised machine learning
- Classification for activity prediction (active / inactive)
- Feature selection (structural features - fingerprints)
- Model validation techniques
- Performance metrics interpretation

**How to perform the analysis:**

1. Data preparation:
  - Define activity threshold to label compounds as active / inactive
  - Select appropriate fingerprint type for molecular encoding
  - Split data into training and testing sets (typically 80/20)
2. Model development:
  - Implement three different ML methods:
    - Random Forest (RF)
    - Support Vector Machines (SVM)
    - Artificial Neural Networks (ANN)
  - Configure appropriate hyperparameters for each model
3. Validation strategy:
  - Perform k-fold cross-validation (typically k=5)
  - Calculate performance metrics for each fold
  - Average results across folds
4. Performance evaluation:
  - Calculate metrics: accuracy, precision, recall, F1 score, ROC-AUC
  - Generate confusion matrices
  - Create ROC curves
5. Feature importance analysis:
  - Extract feature importance from the best-performing model
  - Identify top fingerprint bits that drive predictions
  - Map important bits back to specific molecular fragments

**What to demonstrate:**

- Comparison of model performance with clear justification for model selection

- Analysis of hyperparameter optimization process
- Visualization of model performance (ROC curves)
- Detailed explanation of which structural features (fingerprint bits) were most important for activity prediction
- Visual mapping of important fingerprint bits to molecular substructures
- Critical discussion of model limitations and potential improvements

## **Step 6: Molecular docking (Structure-based virtual screening)**

### **What students need to know:**

- Principles of protein-ligand interactions
- Protein and ligand preparation for docking
- Binding site identification and characterization
- Docking algorithms and scoring functions
- Post-docking analysis techniques

### **How to perform the analysis:**

1. Setup and environment preparation.
2. Protein structure preparation:
  - Import protein structure from PDB / Working directory.
  - Process the structure:
    - Add missing hydrogens
    - Optimize protonation states
    - Fix missing residues and loops
    - Remove crystallographic water molecules (or keep relevant ones)
    - Minimize the structure to relieve steric clashes
3. Ligand preparation:
  - Process the cognate ligand:
    - Generate tautomers and protonation states at target pH
    - Generate 3D conformations
    - Minimize structures
  - Prepare filtered compounds from previous steps:
    - Do the same.
4. Target grid generation:
  - Define the binding site:
    - Using cognate ligand position
  - Set appropriate grid dimensions to encompass binding pocket
  - Configure grid properties (constraints - if needed)
5. Docking process:
  - Dock the cognate ligand first (validation step):
  - Dock the filtered compounds:
6. Basic results analysis:
  - Rank compounds by docking score
  - Visualize binding modes of top-scoring compounds
  - Identify key protein-ligand interactions:
    - Hydrogen bonds

- Hydrophobic contacts
  - $\pi$ - $\pi$  interactions
  - Salt bridges
- 7. Advanced binding site analysis with SiteMap:
  - Characterize binding site properties:
    - Hydrophobic regions
    - Hydrogen bond donor/acceptor regions
  - Identify key residues for ligand binding
- 8. Advanced visualization techniques:
  - Generate protein-ligand interaction diagrams
  - Create surface representations colored by property

**What to demonstrate:**

- Clear explanation of each step in the docking process and its purpose
- Top-scoring compounds with detailed interaction analysis
- Binding site characterization results
- Critical analysis of which structural features contribute to favorable docking scores
- Visual comparison of docking poses between high-scoring compounds and the reference ligand
- Integration of docking results with insights from ligand-based approaches

## Final Presentation Requirements

Students should prepare a 20-minute presentation containing:

1. Introduction (1 minute)
2. Methodology Overview (1-2 minutes)
3. Results and Analysis (10-15 minutes)
4. Conclusions (1-2 minutes)

The presentation should demonstrate:

- Clear understanding of each computational method
- Critical thinking in analyzing and interpreting results
- Ability to integrate insights from multiple computational approaches
- Understanding of the drug discovery process as a whole

Students should be prepared to answer questions about their methodological choices and interpretations of results.

**Target for project:**

1. ChEMBL220 - Acetylcholinesterase. Drug: DONEPEZIL. Mechanism: Acetylcholinesterase inhibitor. Used to treat myasthenia gravis, Ogilvie syndrome, and urinary retention. 6540, <https://pubs.acs.org/doi/10.1021/acs.jmedchem.0c01863>, 7d9p

2. ChEMBL2971 - Janus Kinase 1/2 (JAK1/2). Drug: RUXOLITINIB. Janus Kinase 1/2 (JAK1/2). Mechanism: JAK1/2 inhibitor. JAK-Inhibitoren, die gezielt Immunreaktionen bei rheumatischen Erkrankungen beeinflussen und so die Entzündung hemmen. 5851. <https://pubs.acs.org/doi/10.1021/acs.jmedchem.3c00519> , 8G6Z
3. ChEMBL230 - Cyclooxygenase-2 (COX-2). Drug: Celecoxib. Mechanism: Selective COX-2 inhibitor. 5262, <https://www.nature.com/articles/384644a0> , 1CX2
4. ChEMBL284 - Dipeptidyl Peptidase-4 (DPP-4). Drug: SITAGLIPTIN. Mechanism: DPP-4-Inhibitor. Orales Antidiabetikum aus der Gruppe der DPP-4-Inhibitoren, das zur Behandlung des nicht-insulinpflichtigen Diabetes mellitus vom Typ 2. 4506 <https://onlinelibrary.wiley.com/doi/10.1002/edm2.2> , 1X70
5. ChEMBL206 - Estrogen Receptor Alpha (Nuclear receptor). Drug: Tamoxifen, Raloxifene. 3118. [https://www.cell.com/cell/fulltext/S0092-8674\(00\)81717-1?returnURL=https%3A%2F%2Flinkinghub.elsevier.com%2Fretrieve%2Fpii%2FS0092867400817171%3Fshowall%3Dtrue](https://www.cell.com/cell/fulltext/S0092-8674(00)81717-1?returnURL=https%3A%2F%2Flinkinghub.elsevier.com%2Fretrieve%2Fpii%2FS0092867400817171%3Fshowall%3Dtrue), 3ERT